- Discrete distribution function (d.f.):

  $F_X(x) = P\left(X \le x\right)$ *for all* $x \in R$

  i.e., $F_X(x) = \sum_{x_i \le x} p(x_i)$

- A random variable X which can take on any value (integral as well as fraction) in the interval is called continuous random variable.

- Probability density function (p.d.f.)

  The probability that a random variable X takes a value in the (open or closed) interval $[t_1, t_2]$ is given by the integral of a function called the probability density function $f_X(x)$

  $$P\left(t_1 \le X \le t_2\right) = \int_{t_1}^{t_2} f_X(x)dx$$

  Other names that are used instead of probability density function include density function, continuous probability function, integrating density function.

  Conditions:

  - $f(x) \ge 0 \; \forall \; x$
  - $\int_{-\infty}^{\infty} f(x)dx = 1$

- Continuous distribution function

  If X is a continuous random variable with the probability density function $f_X(x)$, then the function $F_X(x)$ is defined by

  $$F_X(x) = P[X \le x] = \int_{-\infty}^{x} f_X(t)dt, -\infty < x < \infty$$

  is called the distribution function (d.f) or sometimes the cumulative distribution function (c.d.f) of the random variable X.

- Properties of cumulative distribution function (c.d.f.)

  The function $F_X(X)$ or simply $F(X)$ has the following properties

  (i) $0 \le F(x) \le 1, -\infty < x < \infty$

  (ii) $F(-\infty) = \lim_{x \to -\infty} F(x) = 0$, and $F(+\infty) = \lim_{x \to \infty} F(x) = 1$.

  (iii) $F(\cdot)$ is a monotone, non-decreasing function; that is, $F(a) \le F(b)$ for $a < b$.

  (iv) $F(\cdot)$ is continuous from the right; that is, $\lim_{h \to 0} F(x+h) = F(x)$.

  (v) $F'(x) = \frac{d}{dx} F(x) = f(x) \ge 0$

XII Std - Business Maths & Stat EM Chapter 6.indd 145 2/28/2020 2:56:45 PM